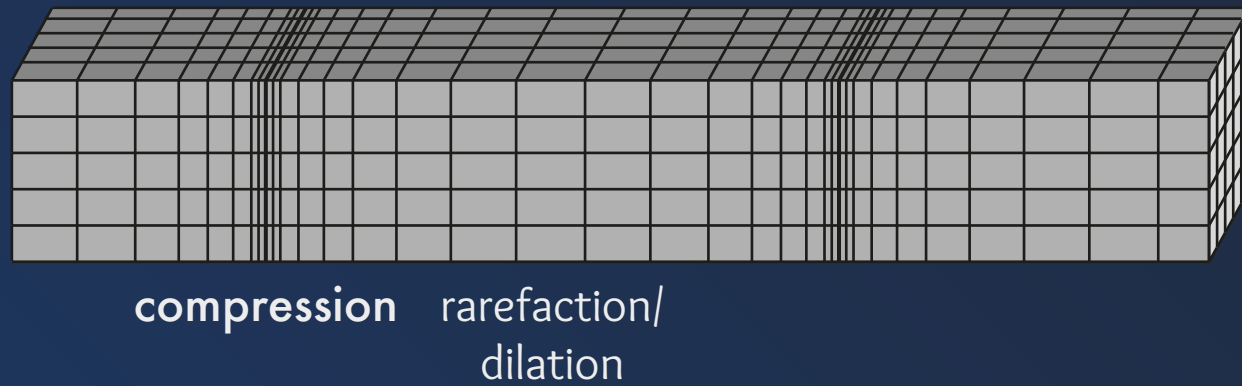


Seismic (Body) Waves

Pressure P-wave



propagation
→
direction

particle
←→
motion

Body waves travel through the interior

P-waves:

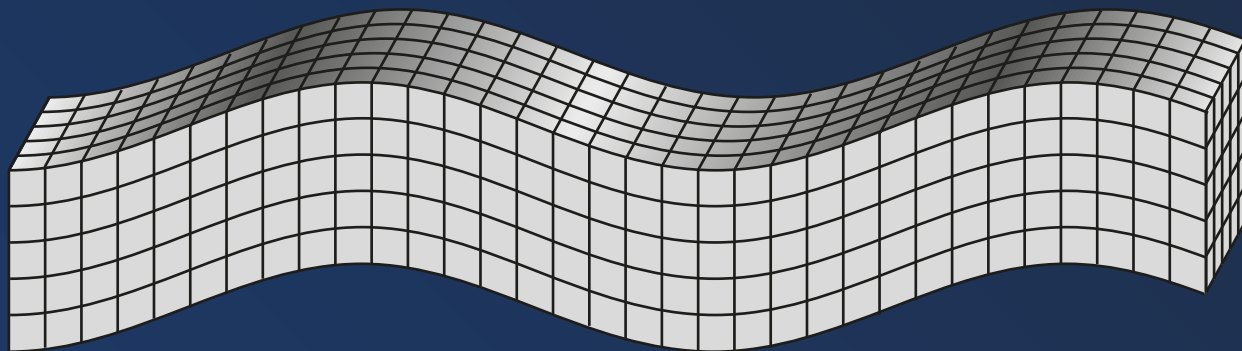
push-pull motion

fast, typically $6-8 \text{ km s}^{-1}$ in crystalline rocks

faster in dense rocks

travels through solids and fluids (including air)

Shear S-wave



propagation
→
direction

SV
SH
particle
motion

S-waves:

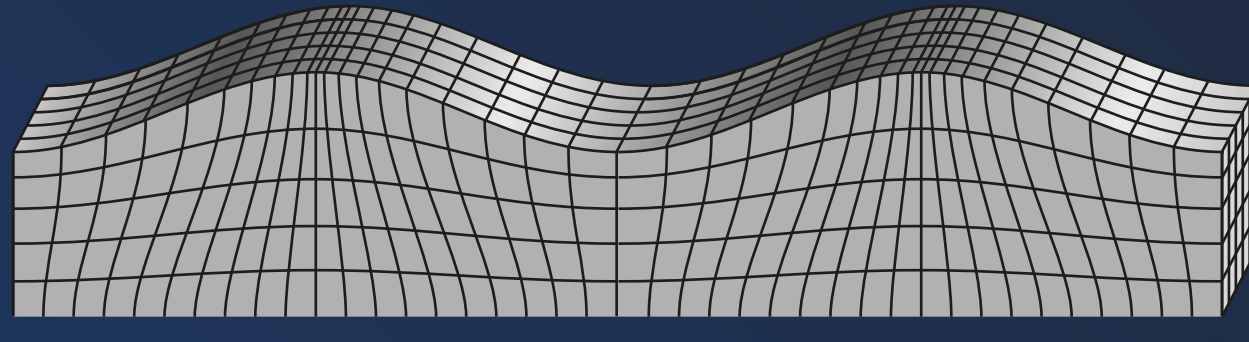
side to side (SH) or up-down (SV) motion

slower than P-waves, typically $3.5-5 \text{ km s}^{-1}$ in crystalline rocks

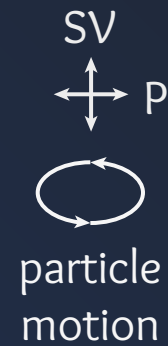
travels only in solids

Seismic (Surface) Waves

Raleigh L_R -wave



propagation
direction

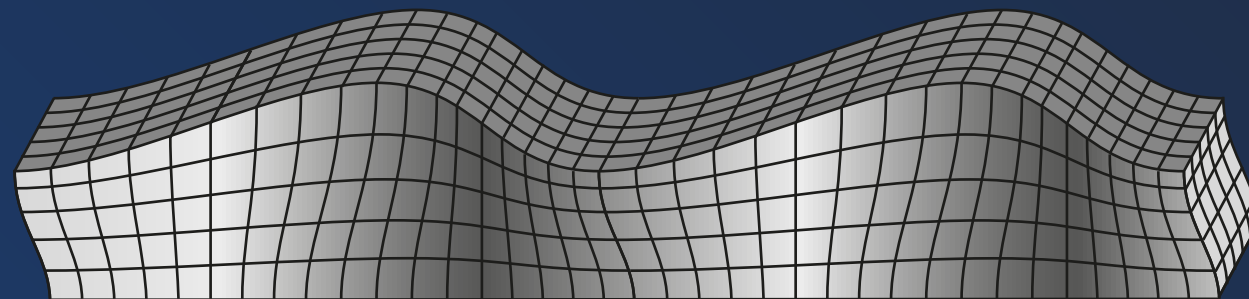


particle
motion

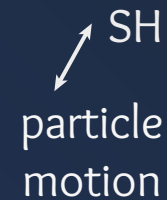
Surface waves do most of the damage during earthquakes

Surface waves:
bound to the surface
amplitude decays with depth
travel slower than body waves

Love L_G -wave



propagation
direction



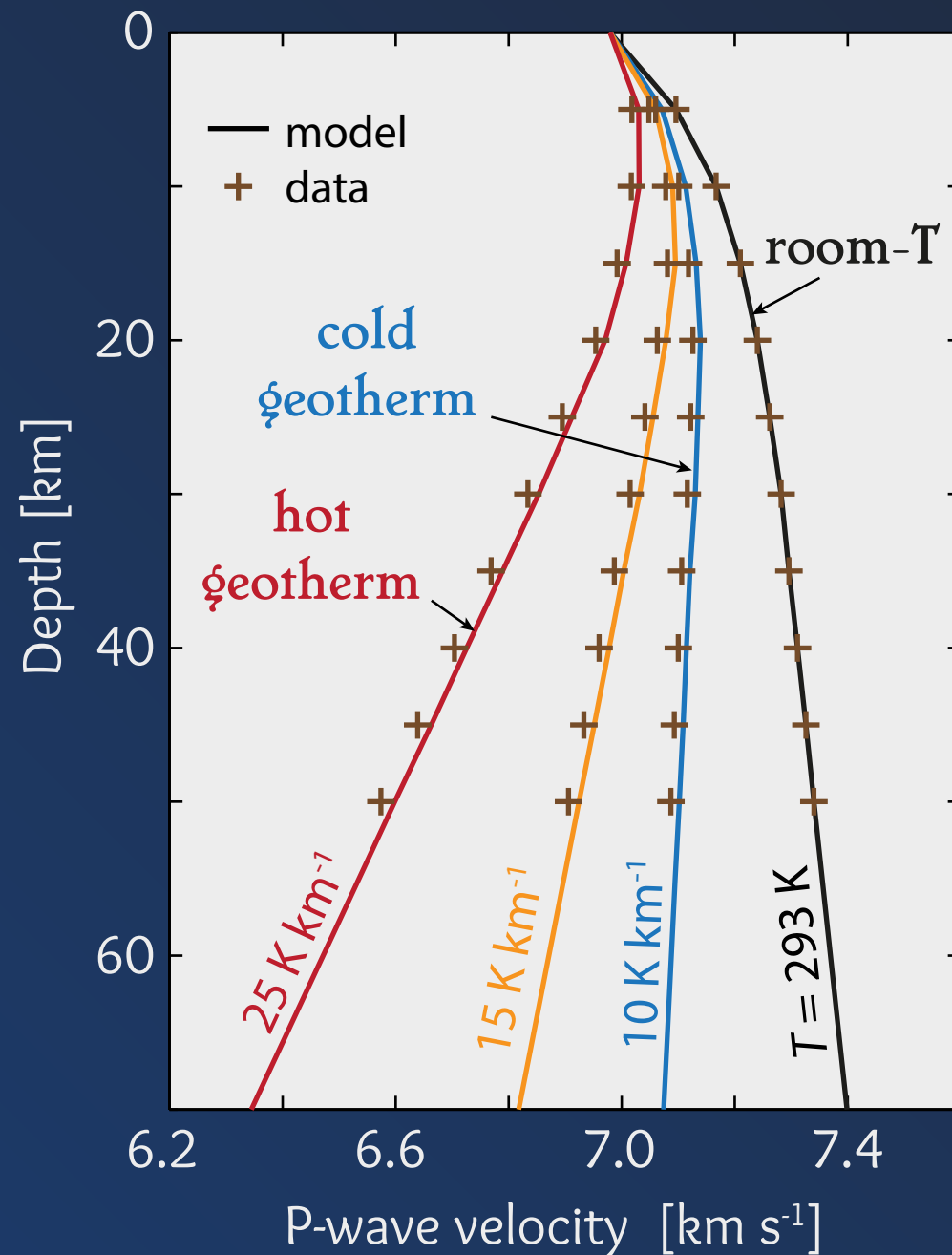
particle
motion

Rayleigh:
vertical shear wave and
compressional wave components

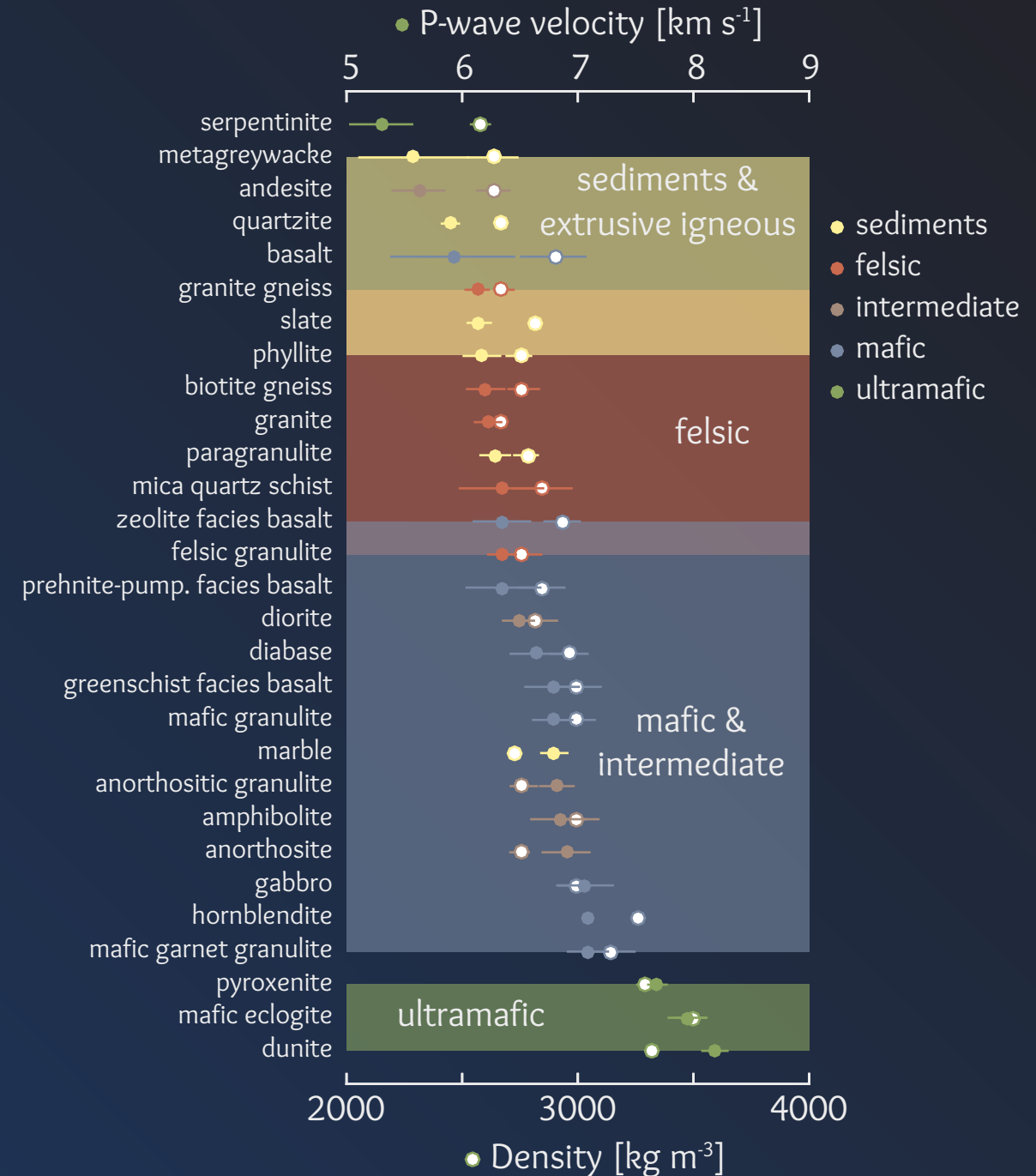
Love:
horizontal shear wave component

Seismic Velocity

Pressure & Temperature Dependence of Seismic Velocity (Gabbro)



Velocity (and density) varies by rock type



Seismograms

Seismograms record the arrival of seismic rays at a location distant from an earthquake

The travel-time (t) of a seismic ray is proportional to the length (x) of the travel path and velocity (v) of the rocks it travels through,

$$t \propto x v^{-1}$$

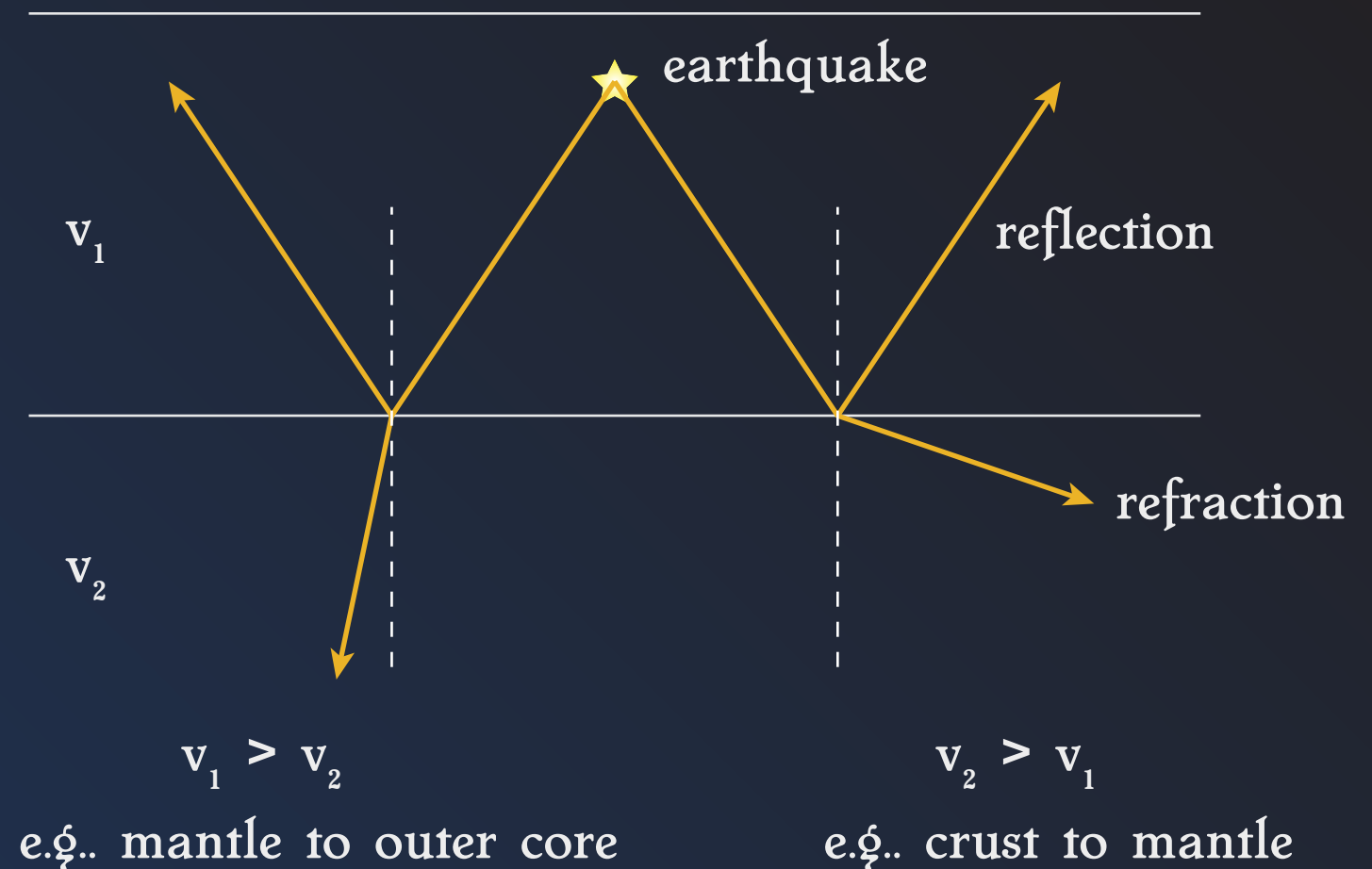
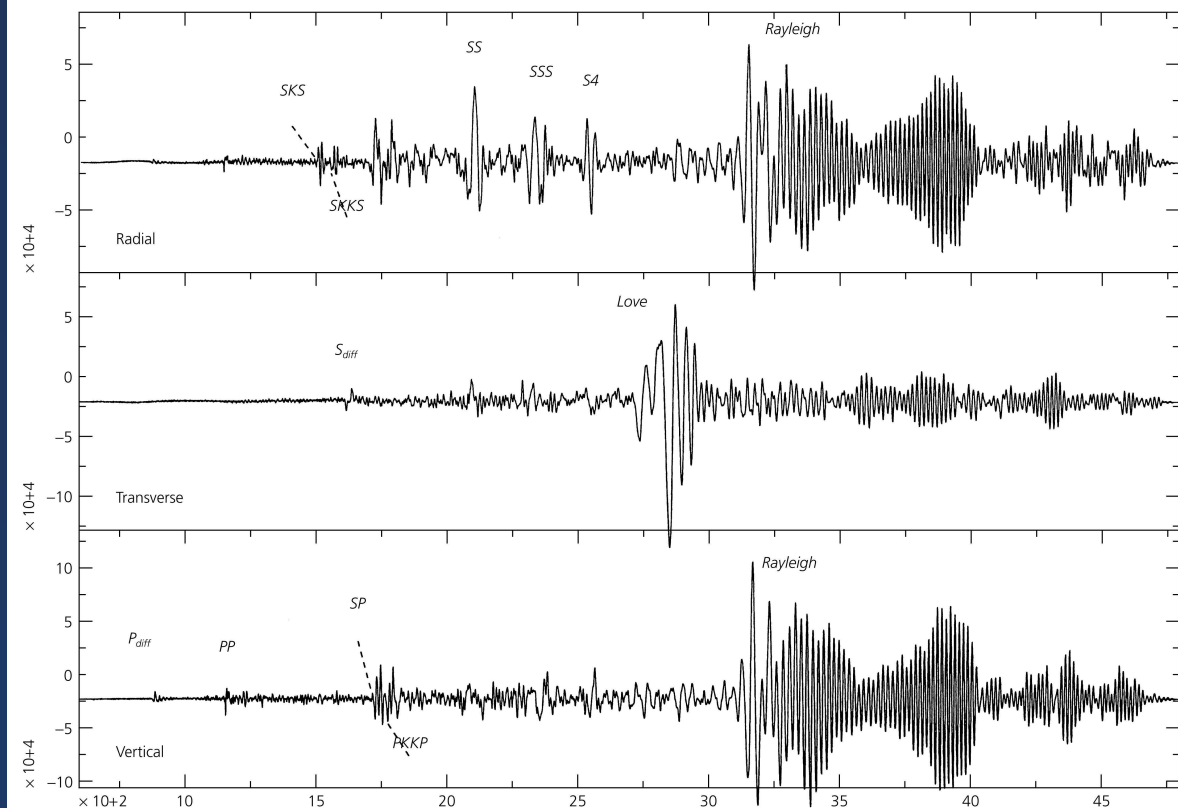


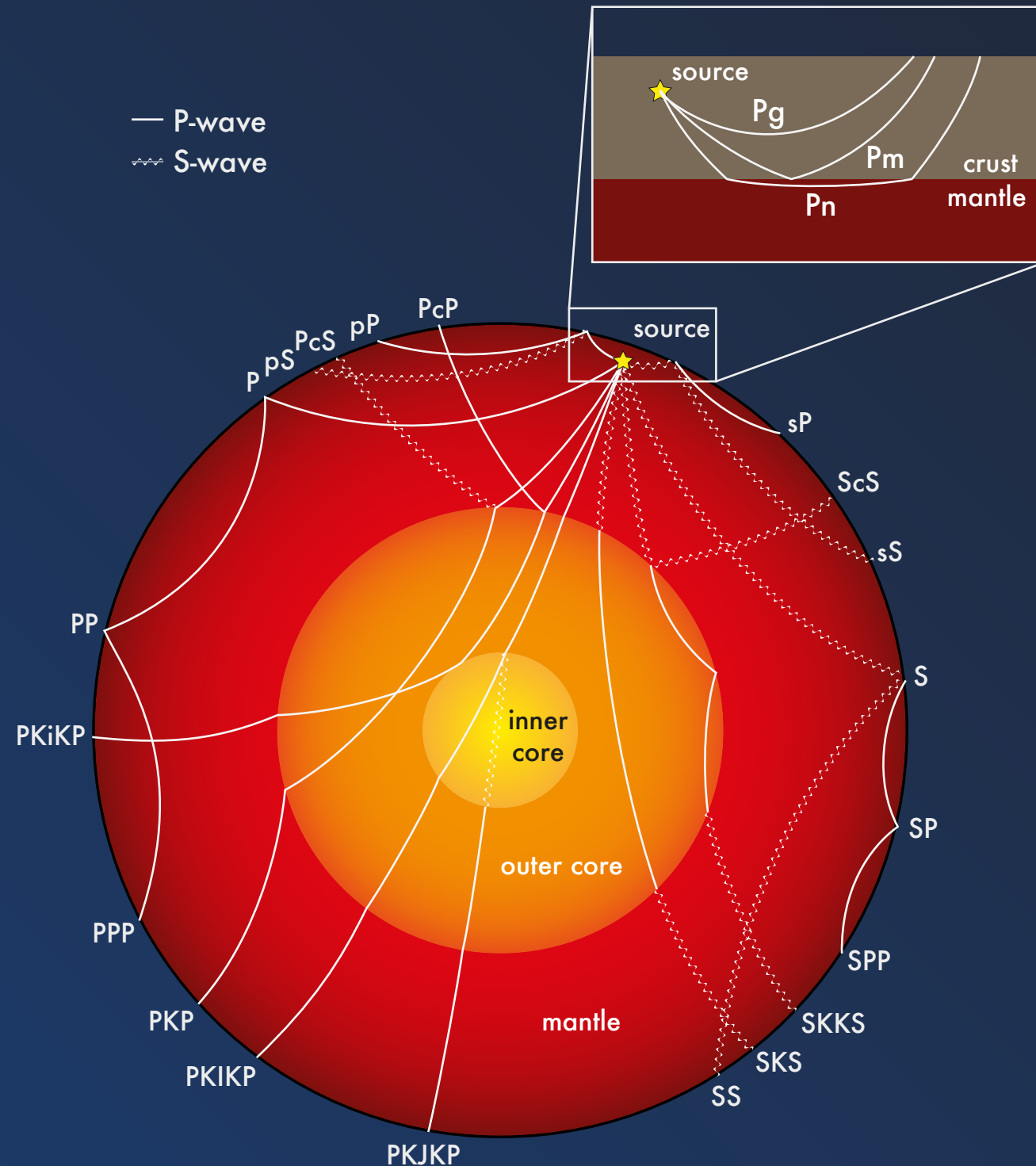
Figure 2.7-1: Seismograms recorded at a distance of 110°, showing surface waves.



Seismic rays are bent (refracted) like light passing through a prism or reflected like light bouncing off a mirror

Refracted waves bend in a predictable way according to Snell's law

Seismic Probing



Seismic waves reflect and refract off structures

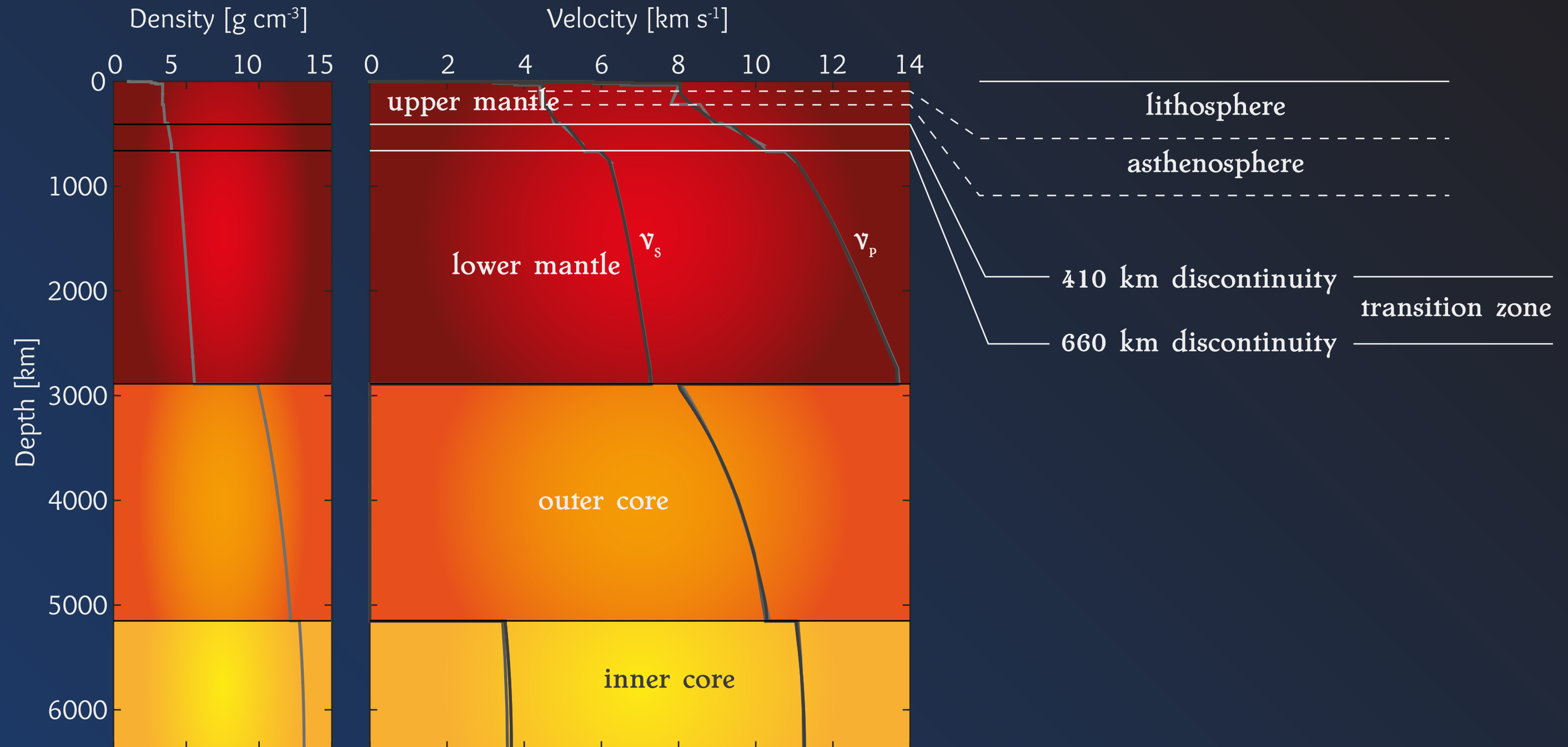
Travel-time is proportional to the length of the ray path

Changes in expected travel-time indicates higher or lower velocities

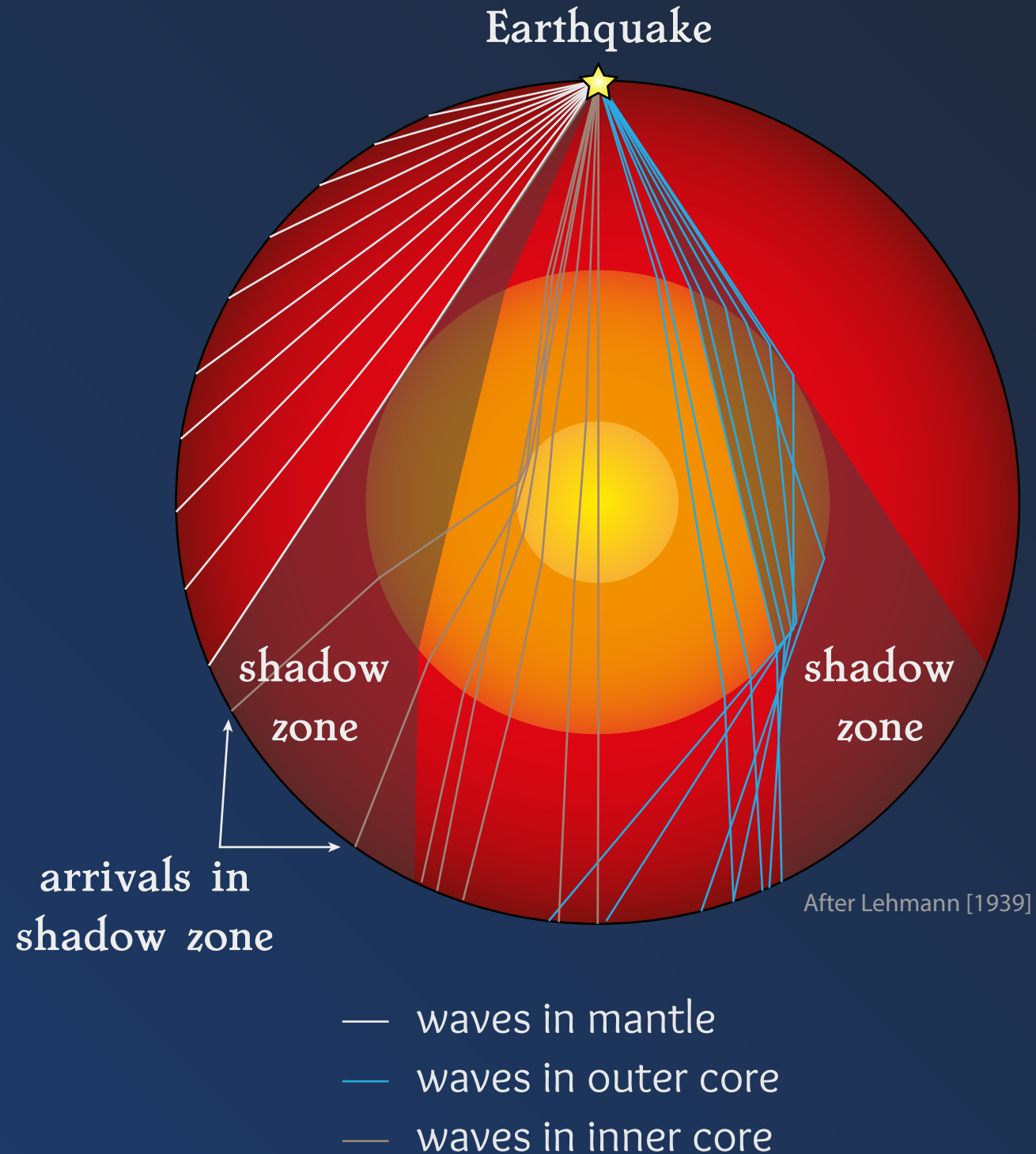
Seismic waves can convert from P to S or S to P as they are reflected off or are transmitted through interfaces

Inner core was discovered by P to S conversions in the inner core

The Seismic View



The Core



Outer core

A shadow zone of P and S waves are created by the outer core, with only P-waves traveling through but leaving a shadow zone where no waves can reach

Discovery of the Inner Core

Arrival of seismic waves at seismometers in the shadow zone created by the outer core indicates a refraction of waves through a faster body within the core (i.e., inner core)

Inge Lehmann, reported the discovery of the inner core in her 1939 paper, P' (shortest title ever?), describing her seismic observations

Inge Lehmann



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